Nicolas Reed

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Education

University of California, Berkeley GPA: 3.63/4.00

B.S. Electrical Engineering & Computer Sciences (EECS) *Expected May 2027*

Coursework: Tapeout, Bringup*, Digital Design and Integrated Circuits (with ASIC Lab), Computer Architecture, Operating Systems, Compilers*, Signals and Circuits, Data Structures, Computer Networking, Data Science

Experience

Apple Cupertino, CA

SoC Design Verification Intern *May 2026 – August 2026*

- Optimized embedded C calibration firmware, cutting image size by over 25% while preserving execution latency.
- Engineered an agentic harness that orchestrates parallel LLM workers to explore firmware improvements.
- Designed a UVM testbench with surrounding hardware models to verify a microcontroller and its firmware.

UC Berkeley Electrical Engineering & Computer Sciences (EECS) Berkeley, CA

Head Teaching Assistant *June 2024 – May 2026*

- Managed a 50+ member course staff supporting 1,200+ students across C, RISC-V, and Python coursework.
- Ranked 1st among staff in helpfulness and teaching effectiveness.
- Authored weekly homework and lab assignments, and maintained course infrastructure using GitHub and Docker.
- Led online forum engagement for the course, answering 1,500+ student questions per semester.

UC Berkeley SLICE Lab Berkeley, CA

Undergraduate Research Assistant *May 2025 – May 2026*

- Ported TVM compilation of ResNet50, TinyLlama, and SmolVLA to Gemmini as a baseline for compiler research.
- Profiled RISC-V BOOM using Top-Down Microarchitecture Analysis counters to identify workload bottlenecks.

Sandisk Milpitas, CA

Software Development Intern *May 2025 – August 2025*

- Reconciled CMDB and inventory data for 40,000+ virtual machines with Pandas to plan a cloud migration.
- Built a LlamaIndex AI assistant to streamline license and contract workflows across 10,000+ applications.

Micross Components - Silicon Turnkey Solutions Milpitas, CA

Engineering Intern *July 2023 – August 2023*

- Conducted mechanical and PCB testing on 1,000+ devices, root-causing compromised semiconductors.

Projects

Physical AI Bringup - *C++, MLIR, C, Python, Chipyard* August 2026 – December 2026

- Implemented an MLIR compiler backend that lowers, schedules, verifies, and encodes custom NPU programs.
- Brought up LLM, VLA, and CNN models end to end on the NPU, offloading nearly all compute bit-exactly.
- Established an NPU kernel library and operator-coverage analysis that maps PyTorch models onto its kernels.

Physical AI Tapeout - *Chisel, Chipyard* December 2025 – May 2026

- Owned RTL implementation of the matrix engine for an Apple-mentored TSMC 16nm SoC tapeout.
- Coordinated block-level interfaces and system connectivity during integration of the NPU accelerator.
- Developed bare-metal functional tests, stress kernels, and VLA workloads for full-chip verification.
- Validated the design on VCU118 and Alveo U250 FPGAs, both directly and via FireSim simulation.

5-Stage Pipelined RISC-V CPU with I/D Caches - *Verilog* October 2025 – December 2025

- Apple NSI (New Silicon Initiative) Design Contest Winner: delivered best ASIC performance and area.
- Boosted throughput and frequency by refining hazard control and forwarding, reducing stall cycles.
- Achieved single-cycle read hits and two-cycle write hits with efficient write-back logic.

Pintos Operating System - *C* August 2025 – December 2025

- Extended core OS support for process control and system calls, handling invalid user memory safely.
- Added multithreading and synchronization primitives, enforcing priority scheduling to prevent inversion.
- Created UNIX-style file system features including caching, file growth, and subdirectories to improve I/O.

Technical Skills

Languages: Verilog, SystemVerilog, C, Python, RISC-V, Chisel, Scala, Java, x86, SQL, OCaml, MATLAB
Tools: Git, Cadence (Indago, Innovus), Synopsys (VCS, Verdi), UVM, GDB, Pandas, NumPy, Docker, LTSpice